Blinded Evaluation of Biological Mechanistic Hypotheses Generated by LLMs

This survey covers **3** biological topics, each containing mechanistic hypotheses generated independently by **3 anonymized LLMs**.
At the beginning of each topic, a **background document summarizes the established dataset context and molecular functions** relevant to the evaluation. Please utilize this information as a common scientific reference if you like and evaluate each hypothesis independently according to the provided criteria.

Step 1 — Hypothesis quality: Score Hypotheses A, B, and C separately on all three pillars:
1. Core biological plausibility — Is the central mechanism compatible with established biological knowledge or reasonable? Evaluate the central causal story as a whole, not every individual detail.

2. Mechanistic specificity and completeness — Does the hypothesis provide a clear and sufficiently complete causal explanation connecting the identified conditions to construct a mechanism?

3. Accuracy of specific molecular claims — Are the stated gene/protein functions, molecular interactions and pathways factually correct and appropriately qualified? If a function is not fully understood by the field, was the uncertainty appropriately communicated.

Use the same 1–5 scale for every pillar:
1 = Fails the criterion: The relevant content is absent, fundamentally flawed, or contradicted.

2 = Weak: The criterion is addressed only minimally and has major gaps, errors, or unsupported elements.

3 = Moderate: The criterion is partially satisfied, but at least one important weakness, uncertainty, or missing connection remains.

4 = Strong: The criterion is largely satisfied, with no major problems and only minor limitations.

5 = Exceptional: The criterion is fully satisfied with no material weakness; this score should be reserved for unusually clear, complete, and well-supported content.

Note: Score each pillar independently. Additional mechanistic detail should improve the specificity score only when it is relevant; inaccurate detail should lower the molecular-accuracy score.

Step 2 — Pairwise agreement: Score pairs of A–B, A–C, and B–C for mechanistic concordance.
1 = No concordance. Different or incompatible mechanisms.
2 = Low concordance. One shared idea, but different causal explanations.
3 = Moderate concordance. Same broad mechanism, but a key role or causal link differs.
4 = High concordance. Same core causal chain; only secondary details differ.
5 = Near-complete concordance. Essentially the same mechanism, roles, directions, and conclusion

Note: Sharing only the features or conditions supplied in the decision trees should receive low score. A score of 4 or 5 requires agreement in the central causal mechanism, not merely the same feature names.

Evaluate only the scientific content of Hypotheses A, B, and C. You may use Google Search to locate scientific sources, but DO NOT use AI chatbots, AI-assisted answer services, or AI-generated summaries.

* Indicates required question

Dataset 1 — Role of Cytoplasmic Actin in EGF-Induced ERK Phosphorylation

First, score Hypotheses A–C separately across the three quality pillars. Then, score the mechanistic agreement of each hypothesis pair before continuing. The background context is attached.

1. Hypothesis A: *

The combined pattern—high cytoplasmic Paxillin, high cytobody Paxillin, and high nuclear YAP—identifies a “Mechanical Priming” phase in which high-density focal adhesions coexist with compartmentalized Paxillin bodies. High cytobody Paxillin represents subcellular segregation of adhesion components into cytoplasmic bodies, likely endocytic vesicles or storage granules containing Paxillin. In this state, the actin cytoskeleton is actively remodeled for high-fidelity force transmission, while moderate nuclear YAP may synergize with the pathway and the cytoskeleton provides the structural scaffold for ERK nuclear entry. EGF induces Paxillin-rich cytoplasmic vesicles that retain active Src/FAK signaling complexes away from degradation, enhancing the pool of substrates available for ERK phosphorylation. Mature focal adhesions amplify MAPK/ERK signaling, while Paxillin trafficking or endosomal recycling is a prerequisite for optimal EGF-induced pERK attribution.

Score this hypothesis on each of the three quality pillars below:

Mark only one oval per row.

	1	2	3	4	5
Core biological plausibility	◯	◯	◯	◯	◯
Mechanistic specificity and completeness	◯	◯	◯	◯	◯
Accuracy of specific molecular claims	◯	◯	◯	◯	◯

2. Hypothesis B: *

The node defines a high-adhesion, high-cytoskeletal-tension, YAP-permissive state in which cytoplasmic Actin strongly predicts pERK: cytoplasmic Paxillin is high, cytobody Paxillin is high, and nuclear YAP is high. Paxillin-rich adhesions provide a scaffold that integrates EGF-EGFR-Ras/Raf/MEK/ERK signaling with Src/FAK-associated adhesion signaling. High cytobody Paxillin represents mature, force-bearing adhesions or central cytoskeletal anchoring sites associated with the strongest Actin-pERK relationship. In this context, cytoplasmic Actin indexes a signaling-competent network that sustains tension, adhesion organization, receptor trafficking, and spatial coordination of EGFR/MAPK signaling. High nuclear YAP marks a mechanotransductively and transcriptionally permissive state that allows EGF-induced ERK activation to be reinforced by actin-paxillin-YAP-associated signaling.

Score this hypothesis on each of the three quality pillars below:

Mark only one oval per row.

	1	2	3	4	5
Core biological plausibility	◯	◯	◯	◯	◯
Mechanistic specificity and completeness	◯	◯	◯	◯	◯
Accuracy of specific molecular claims	◯	◯	◯	◯	◯

3. Hypothesis C: *

This node defines a tense-scaffold phenotype characterized by high global focal-adhesion presence—high cytoplasmic Paxillin and high cytobody Paxillin—and active mechanotransduction marked by high nuclear YAP. High Paxillin indicates that the cell is firmly attached to the substrate, while nuclear YAP indicates that the actin cytoskeleton is under high mechanical tension. Pre-existing integrin-EGFR crosstalk at these focal adhesions uses the tense actin network to cluster and optimally activate the RAF/MEK/ERK cascade after EGF stimulation. EGF-induced pERK activation is strictly dependent on RhoA/ROCK-mediated actin contractility only when whole-cell Paxillin is high. In this primed state, the tension of the actin network, rather than simply the presence of actin filaments, is the driver of pERK.

Score this hypothesis on each of the three quality pillars below:

Mark only one oval per row.

	1	2	3	4	5
Core biological plausibility	◯	◯	◯	◯	◯
Mechanistic specificity and completeness	◯	◯	◯	◯	◯
Accuracy of specific molecular claims	◯	◯	◯	◯	◯

4. Read Hypotheses, then rate each pair for mechanistic concordance. *

Hypothesis A:
The combined pattern—high cytoplasmic Paxillin, high cytobody Paxillin, and high nuclear YAP—identifies a “Mechanical Priming” phase in which high-density focal adhesions coexist with compartmentalized Paxillin bodies. High cytobody Paxillin represents subcellular segregation of adhesion components into cytoplasmic bodies, likely endocytic vesicles or storage granules containing Paxillin. In this state, the actin cytoskeleton is actively remodeled for high-fidelity force transmission, while moderate nuclear YAP may synergize with the pathway and the cytoskeleton provides the structural scaffold for ERK nuclear entry. EGF induces Paxillin-rich cytoplasmic vesicles that retain active Src/FAK signaling complexes away from degradation, enhancing the pool of substrates available for ERK phosphorylation. Mature focal adhesions amplify MAPK/ERK signaling, while Paxillin trafficking or endosomal recycling is a prerequisite for optimal EGF-induced pERK attribution.

Hypothesis B:
The node defines a high-adhesion, high-cytoskeletal-tension, YAP-permissive state in which cytoplasmic Actin strongly predicts pERK: cytoplasmic Paxillin is high, cytobody Paxillin is high, and nuclear YAP is high. Paxillin-rich adhesions provide a scaffold that integrates EGF-EGFR-Ras/Raf/MEK/ERK signaling with Src/FAK-associated adhesion signaling. High cytobody Paxillin represents mature, force-bearing adhesions or central cytoskeletal anchoring sites associated with the strongest Actin-pERK relationship. In this context, cytoplasmic Actin indexes a signaling-competent network that sustains tension, adhesion organization, receptor trafficking, and spatial coordination of EGFR/MAPK signaling. High nuclear YAP marks a mechanotransductively and transcriptionally permissive state that allows EGF-induced ERK activation to be reinforced by actin-paxillin-YAP-associated signaling.

Hypothesis C:
This node defines a tense-scaffold phenotype characterized by high global focal-adhesion presence—high cytoplasmic Paxillin and high cytobody Paxillin—and active mechanotransduction marked by high nuclear YAP. High Paxillin indicates that the cell is firmly attached to the substrate, while nuclear YAP indicates that the actin cytoskeleton is under high mechanical tension. Pre-existing integrin-EGFR crosstalk at these focal adhesions uses the tense actin network to cluster and optimally activate the RAF/MEK/ERK cascade after EGF stimulation. EGF-induced pERK activation is strictly dependent on RhoA/ROCK-mediated actin contractility

only when whole-cell Paxillin is high. In this primed state, the tension of the actin network, rather than simply the presence of actin filaments, is the driver of pERK.

Mark only one oval per row.

	1	2	3	4	5
Hypotheses A and B	⬭	⬭	⬭	⬭	⬭
Hypotheses A and C	⬭	⬭	⬭	⬭	⬭
Hypotheses B and C	⬭	⬭	⬭	⬭	⬭

Dataset 2 — Input Resistance in Developing GABAergic Interneurons

First, score Hypotheses A–C separately across the three quality pillars. Then, score the mechanistic agreement of each hypothesis pair before continuing. The background context is attached.

5. Dataset 2 — Molecular Regulation of Input Resistance in Developing GABAergic Interneurons *

Hypothesis A:
This zone captures a differentiating or early post-mitotic CGE-associated neuromodulatory interneuron state marked by low Coro6, high Cnr1, high Htr3a, and low Flt3. In this state, Fxyd6 modulates the Na+/K+-ATPase, stabilizing ionic gradients and resting membrane properties. High Htr3a creates cation-influx potential and a demand for pump-mediated recovery, while high Cnr1 places the cell in a Gi/o-coupled excitability state in which resting conductances are actively regulated. Low Coro6 and low Flt3 mark cells that have moved away from motility- or progenitor-associated programs, making Fxyd6-mediated ion homeostasis more directly relevant to input resistance. Fxyd6 serves both as a marker of this interneuron state and as a functional regulator of the pump-dependent homeostatic mechanisms that shape or stabilize higher input resistance.

Score this hypothesis on each of the three quality pillars below:

Mark only one oval per row.

	1	2	3	4	5
Core biological plausibility	◯	◯	◯	◯	◯
Mechanistic specificity and completeness	◯	◯	◯	◯	◯
Accuracy of specific molecular claims	◯	◯	◯	◯	◯

6. Dataset 2 — Molecular Regulation of Input Resistance in Developing GABAergic Interneurons *

Hypothesis B:
This node perfectly describes a well-known post-mitotic CGE-derived CCK-positive interneuron population: Coro6 is low, Cnr1 is high, Htr3a is high, and Flt3 is low. Fxyd6 modulates the electrogenic Na+/K+-ATPase, which becomes a primary driver of resting membrane potential and conductance while developing neurons lack the full suite of mature leak-potassium channels. The receptor-pump coupling hypothesis holds that 5-HT3R and CB1R signaling cascades converge to regulate Fxyd6 activity or its interaction with the Na+/K+-ATPase. In this mechanism, CB1R agonism alters input resistance only when Fxyd6 is present, making Fxyd6 a downstream effector of endocannabinoid signaling during development. Low Flt3 marks a post-mitotic, migrating or maturing window in which Fxyd6-mediated input-resistance regulation is transiently required before mature potassium leak channels such as TASK channels take over resting conductance.

Score this hypothesis on each of the three quality pillars below:

Mark only one oval per row.

	1	2	3	4	5
Core biological plausibility	⬭	⬭	⬭	⬭	⬭
Mechanistic specificity and completeness	⬭	⬭	⬭	⬭	⬭
Accuracy of specific molecular claims	⬭	⬭	⬭	⬭	⬭

7. Dataset 2 — Molecular Regulation of Input Resistance in Developing GABAergic Interneurons *

Hypothesis C:
This zone defines developing neurons in which input resistance is maximized and strongly dependent on Fxyd6, with low Coro6, high Cnr1, high Htr3a, and low Flt3 revealing intense neuromodulatory signaling and cytoskeletal quiescence. The scaffold-stabilization hypothesis holds that Fxyd6 acts as a scaffold for Htr3a and Cnr1 heterocomplexes specifically when Coro6-driven actin remodeling is suppressed. Fxyd6 physically links these receptors to the lipid bilayer and prevents their internalization. This stabilization maintains high ligand-gated conductance capacity that paradoxically correlates with high attributed input resistance. This paradox is attributed to the lack of competing Shaker-like potassium leak currents during the same developmental window.

Score this hypothesis on each of the three quality pillars below:

Mark only one oval per row.

	1	2	3	4	5
Core biological plausibility	◯	◯	◯	◯	◯
Mechanistic specificity and completeness	◯	◯	◯	◯	◯
Accuracy of specific molecular claims	◯	◯	◯	◯	◯

8. Dataset 2 — Molecular Regulation of Input Resistance in Developing GABAergic Interneurons *

Read Hypotheses A–C, then rate each pair for mechanistic concordance.

Hypothesis A:
This zone captures a differentiating or early post-mitotic CGE-associated neuromodulatory interneuron state marked by low Coro6, high Cnr1, high Htr3a, and low Flt3. In this state, Fxyd6 modulates the Na+/K+-ATPase, stabilizing ionic gradients and resting membrane properties. High Htr3a creates cation-influx potential and a demand for pump-mediated recovery, while high Cnr1 places the cell in a Gi/o-coupled excitability state in which resting conductances are actively regulated. Low Coro6 and low Flt3 mark cells that have moved away from motility- or progenitor-associated programs, making Fxyd6-mediated ion homeostasis more directly relevant to input resistance. Fxyd6 serves both as a marker of this interneuron state and as a functional regulator of the pump-dependent homeostatic mechanisms that shape or stabilize higher input resistance.

Hypothesis B:
This node perfectly describes a well-known post-mitotic CGE-derived CCK-positive interneuron population: Coro6 is low, Cnr1 is high, Htr3a is high, and Flt3 is low. Fxyd6 modulates the electrogenic Na+/K+-ATPase, which becomes a primary driver of resting membrane potential and conductance while developing neurons lack the full suite of mature leak-potassium channels. The receptor-pump coupling hypothesis holds that 5-HT3R and CB1R signaling cascades converge to regulate Fxyd6 activity or its interaction with the Na+/K+-ATPase. In this mechanism, CB1R agonism alters input resistance only when Fxyd6 is present, making Fxyd6 a downstream effector of endocannabinoid signaling during development. Low Flt3 marks a post-mitotic, migrating or maturing window in which Fxyd6-mediated input-resistance regulation is transiently required before mature potassium leak channels such as TASK channels take over resting conductance.

Hypothesis C:
This zone defines developing neurons in which input resistance is maximized and strongly dependent on Fxyd6, with low Coro6, high Cnr1, high Htr3a, and low Flt3 revealing intense neuromodulatory signaling and cytoskeletal quiescence. The scaffold-stabilization hypothesis holds that Fxyd6 acts as a scaffold for Htr3a and Cnr1 heterocomplexes specifically when Coro6-driven actin remodeling is suppressed. Fxyd6 physically links these receptors to the lipid bilayer and prevents their internalization. This stabilization maintains high ligand-gated conductance

capacity that paradoxically correlates with high attributed input resistance. This paradox is attributed to the lack of competing Shaker-like potassium leak currents during the same developmental window.

Mark only one oval per row.

	1	2	3	4	5
Hypotheses A and B	⬭	⬭	⬭	⬭	⬭
Hypotheses A and C	⬭	⬭	⬭	⬭	⬭
Hypotheses B and C	⬭	⬭	⬭	⬭	⬭

Dataset 3 — PTMA-Centered Regulation of Ki-67 Expression in Breast Cancer

First, score Hypotheses A–C separately across the three quality pillars. Then, score the mechanistic agreement of each hypothesis pair before submitting. The background context is attached.

9. Dataset 3 — PTMA-Centered Regulation of Ki-67 Expression in Breast Cancer *

Hypothesis A:
The “Hyper-Translational Mitotic” phenotype appears when HNRNPA2B1, TUBB4B, and RPLP0 are all high, creating a transcriptomic microenvironment dominated by RNA processing, ribosome biogenesis, and mitotic structural components. PTMA acts as an upstream chromatin remodeler that drives proliferation-gene transcription, but its effect culminates in Ki-67 expression only when HNRNPA2B1 supplies RNA-processing machinery and RPLP0 supplies massive translational capacity. TUBB4B provides the mitotic-spindle building block required for successful cell division. In this state, PTMA is the ignition, and the HNRNPA2B1/RPLP0/TUBB4B axis is the engine required for successful mitosis in breast cancer cells. PTMA-mediated chromatin unwinding may open the HNRNPA2B1 and RPLP0 promoters, while HNRNPA2B1 may regulate the splicing or export of TUBB4B and MKI67 transcripts.

Score this hypothesis on each of the three quality pillars below:

Mark only one oval per row.

	1	2	3	4	5
Core biological plausibility	◯	◯	◯	◯	◯
Mechanistic specificity and completeness	◯	◯	◯	◯	◯
Accuracy of specific molecular claims	◯	◯	◯	◯	◯

10. Dataset 3 — PTMA-Centered Regulation of Ki-67 Expression in Breast Cancer *

Hypothesis B:
The co-expression of high HNRNPA2B1, high TUBB4B, and high RPLP0 defines a state of maximal cellular plasticity, with simultaneous splicing, translation, and microtubule-polymerization activity required for rapid division. PTMA acts as a master regulator, potentially a scaffolding protein or RNA-binding element, that stabilizes the co-transcription of splicing factors and cytoskeletal components. High HNRNPA2B1 and RPLP0 couple intense pre-mRNA processing and transcript maturation to immediate import into translating ribosomes. High TUBB4B supplies beta-tubulin for stress-fiber formation and neuronal or epithelial structural integrity before mitosis. PTMA may bind HNRNPA2B1 or TUBB4B promoters or 3-prime untranslated regions to enhance co-transcription, while HNRNPA2B1 may stabilize TUBB4B mRNA.

Score this hypothesis on each of the three quality pillars below:

Mark only one oval per row.

	1	2	3	4	5
Core biological plausibility	◯	◯	◯	◯	◯
Mechanistic specificity and completeness	◯	◯	◯	◯	◯
Accuracy of specific molecular claims	◯	◯	◯	◯	◯

11. Dataset 3 — PTMA-Centered Regulation of Ki-67 Expression in Breast Cancer *

Hypothesis C:
The Node signature isolates a breast cancer state with three coordinated modules: high HNRNPA2B1-associated RNA processing, high TUBB4B-associated microtubule and mitotic execution, and high RPLP0-associated ribosomal and translation capacity. Within this state, PTMA acts upstream as a chromatin-associated proliferation amplifier that opens or sustains transcriptionally permissive chromatin at MKI67 and other cell-cycle loci. HNRNPA2B1 then supports the processing, export, stability, or splicing of the resulting proliferation-associated transcripts. RPLP0 supplies the ribosomal capacity for protein synthesis, while TUBB4B supplies the microtubule and mitotic machinery needed to execute the proliferative program. Together, these components form a fully enabled proliferation circuit in which transcriptional activation, RNA maturation, translation, and mitotic competence are simultaneously high, stabilizing Ki-67 expression and the cycling state.

Score this hypothesis on each of the three quality pillars below:

Mark only one oval per row.

	1	2	3	4	5
Core biological plausibility	◯	◯	◯	◯	◯
Mechanistic specificity and completeness	◯	◯	◯	◯	◯
Accuracy of specific molecular claims	◯	◯	◯	◯	◯

12. Dataset 3 — PTMA-Centered Regulation of Ki-67 Expression in Breast Cancer *

Read Hypotheses A–C, then rate each pair for mechanistic concordance.

Hypothesis A:
The “Hyper-Translational Mitotic” phenotype appears when HNRNPA2B1, TUBB4B, and RPLP0 are all high, creating a transcriptomic microenvironment dominated by RNA processing, ribosome biogenesis, and mitotic structural components. PTMA acts as an upstream chromatin remodeler that drives proliferation-gene transcription, but its effect culminates in Ki-67 expression only when HNRNPA2B1 supplies RNA-processing machinery and RPLP0 supplies massive translational capacity. TUBB4B provides the mitotic-spindle building block required for successful cell division. In this state, PTMA is the ignition, and the HNRNPA2B1/RPLP0/TUBB4B axis is the engine required for successful mitosis in breast cancer cells. PTMA-mediated chromatin unwinding may open the HNRNPA2B1 and RPLP0 promoters, while HNRNPA2B1 may regulate the splicing or export of TUBB4B and MKI67 transcripts.

Hypothesis B:
The co-expression of high HNRNPA2B1, high TUBB4B, and high RPLP0 defines a state of maximal cellular plasticity, with simultaneous splicing, translation, and microtubule-polymerization activity required for rapid division. PTMA acts as a master regulator, potentially a scaffolding protein or RNA-binding element, that stabilizes the co-transcription of splicing factors and cytoskeletal components. High HNRNPA2B1 and RPLP0 couple intense pre-mRNA processing and transcript maturation to immediate import into translating ribosomes. High TUBB4B supplies beta-tubulin for stress-fiber formation and neuronal or epithelial structural integrity before mitosis. PTMA may bind HNRNPA2B1 or TUBB4B promoters or 3-prime untranslated regions to enhance co-transcription, while HNRNPA2B1 may stabilize TUBB4B mRNA.

Hypothesis C:
The node signature isolates a breast cancer state with three coordinated modules: high HNRNPA2B1-associated RNA processing, high TUBB4B-associated microtubule and mitotic execution, and high RPLP0-associated ribosomal and translation capacity. Within this state, PTMA acts upstream as a chromatin-associated proliferation amplifier that opens or sustains transcriptionally permissive chromatin at MKI67 and other cell-cycle loci. HNRNPA2B1 then supports the processing, export, stability, or splicing of the resulting proliferation-associated transcripts. RPLP0 supplies the ribosomal

capacity for protein synthesis, while TUBB4B supplies the microtubule and mitotic machinery needed to execute the proliferative program. Together, these components form a fully enabled proliferation circuit in which transcriptional activation, RNA maturation, translation, and mitotic competence are simultaneously high, stabilizing Ki-67 expression and the cycling state.

Mark only one oval per row.

	1	2	3	4	5
Hypotheses A and B	◯	◯	◯	◯	◯
Hypotheses A and C	◯	◯	◯	◯	◯
Hypotheses B and C	◯	◯	◯	◯	◯

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